CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
5th Semester of B. Pharm. Examination
University Theory Examination November/December 2016
PH316 Medicinal Chemistry-I

Date: 21.11.2016 , Monday Time: 10:00 a.m. to 01:00 p.m. Maximum Marks: 80

Instructions:

1. **There are three sections in this question paper.**
2. **SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.**
3. **Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.**
4. **Figures to right indicate marks.**
5. **Draw neat labeled sketches wherever necessary.**

SECTION - I

Q 1 Attempt all questions. Each question is of one mark. 20

1. Identify which of the following structural features is present in many muscarinic antagonists but not in muscarinic agonists.
[A] An ester
[B] A quaternary nitrogen
[C] An acyl group containing one or two aromatic rings
[D] An ethylene bridge between an ester and a quaternary nitrogen
2. Identify the tactic which is used to obtain Beta adrenoceptor selectivity for adrenergic agonists.
[A] Introduction of a naphthalene ring
[B] Introduction of a bulky N-alkyl substituent
[C] Removal of one of the catechol OH group
[D] Addition of an Alpha-methyl group (alpha to the carbon bearing the amine)
3. Which of the following is a clinical use for a nicotinic agonist?
[A] Treatment of myasthenia gravis
[B] Switching on the gastrointestinal tract
[C] Switching on the urinary tract
[D] Decreasing heart muscle activity in certain heart defects
4. Which of the following is a general characteristic of non-depoarizing neuromuscular agents?
[A] The alkyl groups on nitrogen is methyl.
[B] Two positively charged centres are required at a specific distance apart.
[C] Large branched acyl groups are good for activity.
[D] The nitrogen is primary in nature..
5. Which compounds is arylethanolamines derivatives.
[A] Sotalol
[B] Propranolol
[C] Acebutolol
[D] Atenol

6. For a H_2-Receptor antagonist, _________ Heterocycle ring is responsible for activity.
[A] Imidazole
[B] Thiophene
[C] Furan
[D] All

7. Basic pharmacophore for Proton Pump Inhibitors is
[A] Piperazinylmethylbenzimidazole
[B] pyridylmethylimidazol
[C] pyridylmethylsulfinylbenzimidazole
[D] Pyridylmethylphospobenzimidazole

8. ________________________ is not directly acting cholinergic drugs.
[A] Physostigmine
[B] Acetylcholine
[C] Carbachol
[D] Pilocarpine

9. Azo compounds are metabolize by ______________ and convert into amine
[A] Hydrolysis
[B] Reduction
[C] Oxidation
[D] Conjugation

10. Following feature reduce activity of 1st generation H_1 Antagonist.
[A] Diary ring system
[B]Tertiary amine
[C] Ethelene side bridge between X and N.
[D] Branching of ethylene chain.

11. Metabolism of phenacetin produce metabolite which is responcible for methemoglobinamia.
[A] Phenitidine
[B] N- Hydroxyphenacetin
[C] N-Methooxyphenatidine
[D] N-Methoxyphenacetin

12. Which of the following is π(pi)-Deficient ring system?
[A] Pyrrole
[B] Pyridine
[C] Thiophene
[D] Benzene

13. Reactivity order towards electrophilic aromatic substitution reaction in pyrrole, pyridine and benzene.
[A] Pyrrole > Benzene > Pyridine
[B] Pyridine >Pyrrole > Benzene
[C] Pyridine > Benzene >Pyrrole
[D] Pyridine >Pyrrole > Benzene

14. All statements are true about the Five member heterocyclic ring system contains one hetero atom; Except ONE
[A] All are aromatic
[B] Nucleophilic Substitution is Easy
[C] Carbon is Sp2 hybridised
[D] Lone pair electrons on hetero atom is in p-orbital so it is overlaps with the carbon porbital

15. In which ring system, lone pair of electorns are not involved in aromaticity.
[A] Pyrrole
[B] Thiophene
[C] Furan
[D] Pyridine

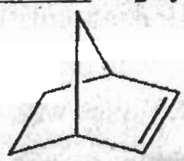
16. ____________________ is an Inhibitor of Mediator Release (Mast cell Stabilizer)
[A] Monteleukast
[B] Hydrocortisone
[C] Cromolyn sodium
[D] Ipratropium

17. Bischler-Napieralski Reaction is a method of preparation of ____________ heterocycle.
[A] Quinoline
[B] Indole
[C] Isoindole
[D] Isoquinoline

18. ______________ is known as performance enahancing drugs in athlets.
[A] Salbutamol
[B] Clenbuterol
[C] Fenoterol
[D] Indacaterol

19. In Indole, S_EAr attack at _______ position is more stable.
[A] C-2 Position
[B] C-3 Position
[C] C-4 Position
[D] C-5 Position

20. Following is an example of _________ ring system.



[A] Bridgehead System
[B] Fused Heterocycles
[C] Heterocyclic Assemblies
[D] Spirocyclic System

SECTION – II

Q 2	Attempt any **TWO** of the following	
A	Define drug metabolism. Describe any four drug oxidation reaction with examples.	05
B	Write a structre activity relationship (SAR) of para sympatomimetics.	05
C	Write a mechanism of Paal Knorr Synthesis of Pyrrole.	05
Q 3	Attempt any **TWO** of the following	
A	Diagramtic presentation of Cytochrome P-450 catalyze metabolism with differentb steps invold in it.	05
B	Chemical Classification of parasympatholytics agents with atleast one structure from each class.	05
C	Classify Antiashtamatic Agent with atleast one structure form each class.	05

SECTION – III

Q 4	Attempt any **FOUR** of the following	
A	Enlist type of drug receptor interaction. Explain signal transduction in brief.	05
B	Explain SAR of Sympatholytic agents.	05
C	Give chemical classification of sympathomimetics agents with their structure.	05
D	Explain SAR of anti allergic (H_1 Antagonist) agents.	05
E	Explain Fisher Indole Synthesis with mechanism.	05
F	Explain Skarup Synthesis of Quinoline with their mechanism.	05
Q 5	Attempt any **FOUR** of the following	
A	Role of Medicinal Chemist in the Modern Drug Discovery and Development	05
B	Write a short note on proton pump inhibitors.	05
C	Write a short note on anti cholinesterase.	05
D	Write any one method of preparation of Imidazole and Oxazole.	05
E	Write a mechanism of Hantzsch Pyridine Synthesis.	05
F	Give synthetic pathway of following drugs: 1.Pentaprazole 2. Salbutamol	05